A. In a RealNVP flow model, given an input vector **x**=[2 3] ^{T},

i) what will be the value of transformed output vector y?

ii) Calculate the log determinant of the Jacobian.

Given an input vector $\mathbf{x} = (x_1, x_2)$, the output vector $\mathbf{y} = (y_1, y_2)$ is computed by:

$$y_1 = x_1$$
$$y_2 = x_2 \cdot \exp(s(x_1)) + t(x_1)$$

Where $s(x_1)$ and $t(x_1)$ are scale and translation functions, typically parameterized by neural networks.

Suppose in a specific RealNVP layer:

- $s(x_1) = 0.5x_1$
- $t(x_1) = 2x_1 + 1$

B. What type of masking patterns are used in RealNVP for implementing variable partitioning? What is the purpose of masking in RealNVP?

[2+3+2= 7 Marks]

Q: Given $x = [2 \quad 3]^T$ $\boxed{\begin{matrix} x_1 = 2 \\ x_2 = 3 \end{matrix}}$

$$y_1 = x_1 \qquad y_2 = x_2 \cdot \exp(s(x_1)) + t(x_1)$$

$$s(x_1) = 0.5x_1 \qquad t(x_1) = 2x_1 + 1$$

A (i) Compute transformed output y

1. $s(x_1) = 0.5 \times 2 = 1$
2. $t(x_1) = 2 \times 2 + 1 = 5$
3. plug for y_2

$$y_2 = x_2 \cdot e^{s(x_1) + t(x_1)}$$

$$= 3e^1 + 5 = 3e + 5$$

4. y_1 is unchanged : $y_1 = x_1 = 2$

So, $\boxed{y = \begin{bmatrix} 2 \\ 3e+5 \end{bmatrix} = \begin{bmatrix} 2 \\ 13.15 \end{bmatrix}}$ Ans.

A(ii) Log determinant of Jacobian

$$y = f(x) \qquad J = \frac{\partial y}{\partial x}$$

$$J = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{bmatrix}$$

Derivative w.r.t x_2

$$* \; y_1 = x_1 \Rightarrow \boxed{\frac{\partial y_1}{\partial x_1} = 1}$$

$$\boxed{\frac{\partial y_1}{\partial x_2} = 0}$$

$$* \; y_2 = x_2 \exp(s(x_1)) + t(x_1)$$

$$\boxed{\frac{\partial y_2}{\partial x_2} = \exp(s(x_1)) = \exp(0.5x_1)}$$

Derivative w.r.t x_1

$$\boxed{\frac{\partial y_1}{\partial x_1} = 1} \qquad * \; \frac{\partial y_2}{\partial x_1} = x_2 \exp(s(x_1)) \cdot s'(x_1) + t'(x_1)$$

$$= x_2 \exp(s(x_1)) \cdot (0.5) + 2$$

$$\boxed{t'(x_1) = 2} \qquad \boxed{s'(x_1) = 0.5}$$

Putting these into J.

$$J = \begin{bmatrix} 1 & 0 \\ x_2 \exp(s(x_1)) \cdot (0.5) + 2 & \exp(s(x_1)) \end{bmatrix}$$

$\Rightarrow 3 \exp(0.5 \times 2) \times 0.5 + 2$

$\Rightarrow 3 \exp(1) * 0.5 + 2$

$\Rightarrow \boxed{1.5e + 2}$

$\Rightarrow \exp(0.5 \times 2)$

$\Rightarrow \exp(1)$

$\boxed{\Rightarrow e}$

$$J = \begin{bmatrix} 1 & 0 \\ 1.5e+2 & e \end{bmatrix}$$

$\det(J) = 1 \cdot e - 0(1.5e+2) = e$

$\boxed{\log(\det(J)) = \log(e) = 1}$ Ans.

B. Masking Patterns

* RealNVP uses binary masks to partition input vector in 2 parts: one part remains unchanged, & other is transformed.

* Common patterns.

Alternating masking = [1, 0, 1, 0, ...]

Purpose of Masking

* Enables invertibility of transformation.
* Allows efficient computation of Jacobian determinant
* Facilitates learning complex distributions by stacking masked layers.